

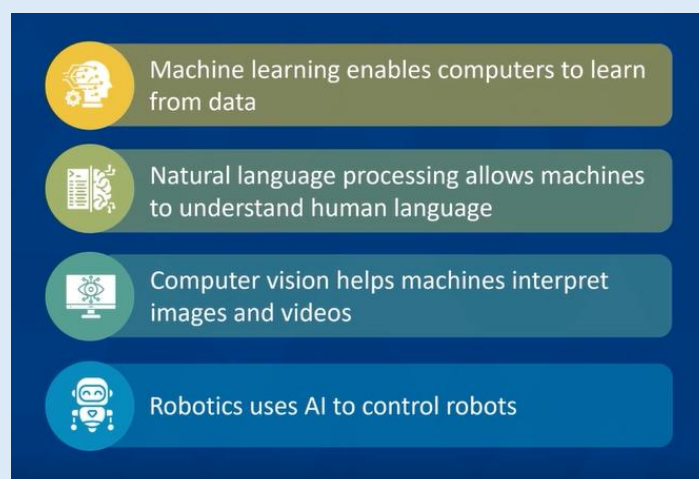
Module 13: Introduction to Artificial Intelligence and Machine Learning

Module Overview

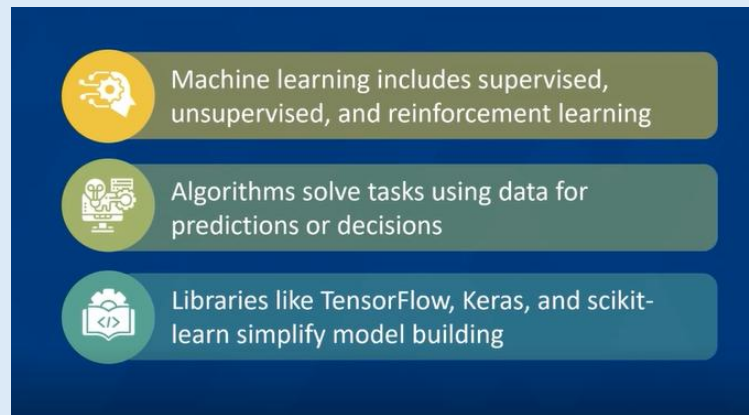
- Artificial Intelligence refers to building systems that mimic human intelligence, enabling machines to learn, understand, and make decisions.



- AI encompasses various subfields including machine learning, natural language processing, computer vision, and robotics, each addressing different intelligent tasks.



- Recent AI advancements include game-playing champions, self-driving cars, disease diagnosis, personalised education, and AI-generated creative content.
- Machine learning, a core subfield of AI, includes supervised, unsupervised, and reinforcement learning, each solving different data-driven problems.
- Popular Python libraries like TensorFlow, Keras, and scikit-learn simplify building and training machine learning models with ready-made tools.

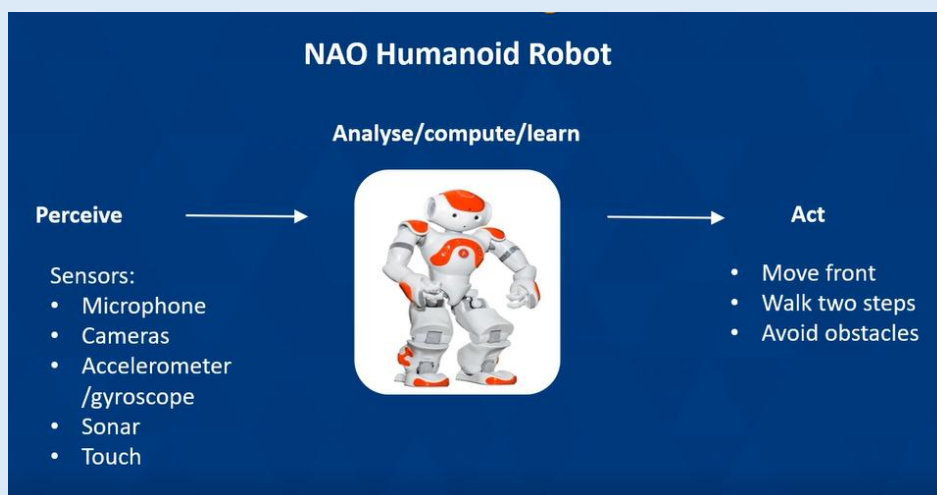


What is Artificial Intelligence (AI)?

- Human intelligence operates in three core steps: perceiving through senses, processing information to learn or decide, and performing actions.



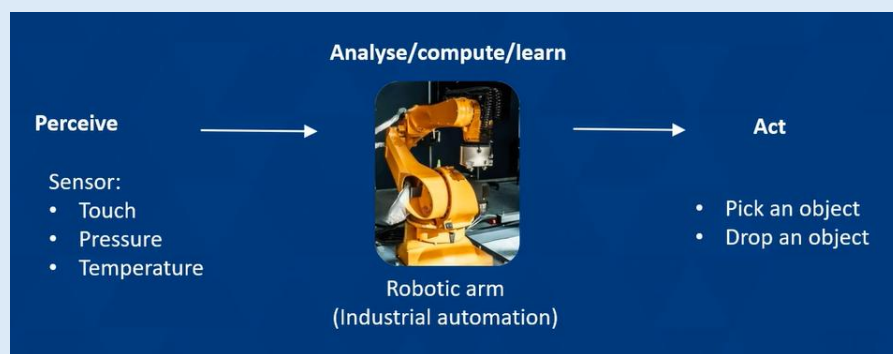
- AI systems function similarly by perceiving inputs through sensors, analysing data to make decisions, and acting based on outcomes.



- Real-world AI applications include Siri for voice commands, Google Maps for route planning, and self-driving cars for automated navigation.



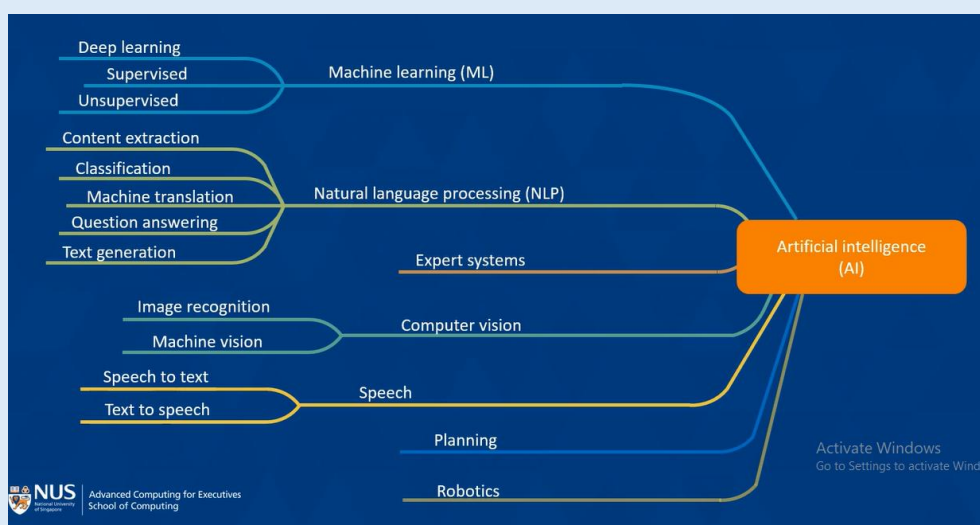
- Gmail, robotic arms, and social media platforms use AI to streamline tasks, improve efficiency, and personalise user experiences.



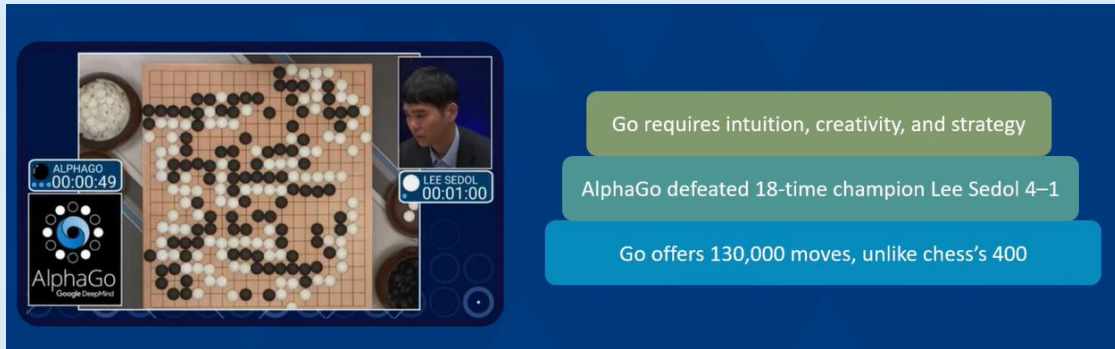
- AI fundamentally simulates human-like intelligence by sensing data, learning or computing from it, and performing context-based actions.

Branches of AI

- Artificial Intelligence includes diverse fields like machine learning, NLP, expert systems, vision, speech processing, planning, and robotics, each with practical real-world applications.



- Deep learning, a part of machine learning, powers systems like AlphaGo that perform strategic thinking and decision-making in complex environments.



- Natural language processing enables human-computer interaction through systems like ChatGPT, Copilot, and Gemini, enhancing dialogue, translation, and code generation.

- NLP focuses on human-computer interaction through natural languages.
- Key NLP tasks are text generation, machine translation, question answering, and content extraction.
- LLMs enhance different facets of human-computer interaction.
- Examples include:
 - ChatGPT – Engages in dialogue and provides information.
 - Copilot – Assists developers with AI-driven code suggestions.
 - Gemini – Aims for broader multi-modal capabilities, extending beyond text.

- Computer vision and speech recognition help machines interpret images and voice, enabling applications like facial recognition, industrial automation, and smart speakers.

AI Vision enables machines to interpret and understand visual data.

Object recognition widely used in applications such as:

- Facial recognition
- Autonomous vehicles
- Security systems

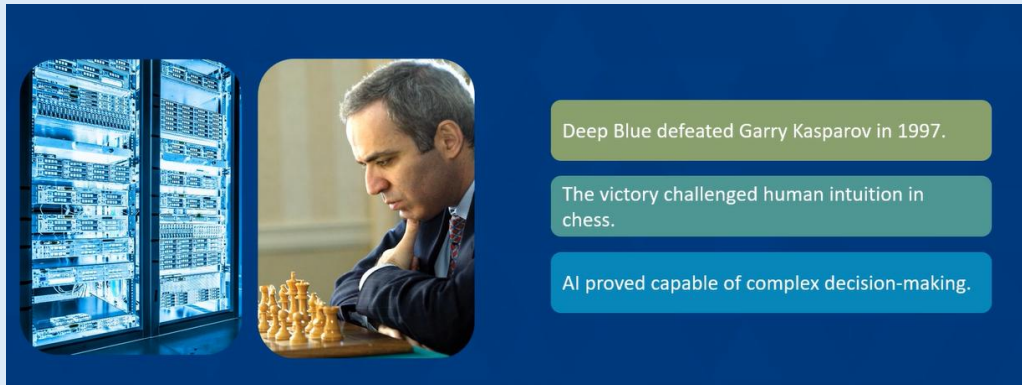
Machine vision enhances industrial automation by:

- Detecting defects
- Guiding robotics
- Ensuring quality control

- Robotics combines path planning and AI coordination to enable autonomous navigation and intelligent surveillance, supporting systems like delivery bots and PTZ camera networks.

Landmark Achievements in AI

- Deep Blue's 1997 victory over chess champion Garry Kasparov marked AI's ability to handle complex strategic decision-making.



- The 2005 DARPA Grand Challenge proved the potential of autonomous vehicles, with AI-controlled cars navigating a 212 km desert course.



- IBM Watson's 2011 win on Jeopardy highlighted AI's growing strength in natural language understanding and fast information retrieval.



The image shows the interface of the Go match between AlphaGo and Lee Sedol. It features three boards with scores: \$24,000, \$77,147, and \$21,600. The boards display questions like 'Who is stronger?' and 'Who is the best player?' with corresponding scores.

- Defeated top champions using advanced natural language processing
- Demonstrated AI's ability to understand, analyse, and respond to complex questions
- Showcased AI's potential in quick information retrieval

- AlphaGo, OpenAI Five, and AlphaStar demonstrated AI's capacity to master complex, strategic games involving real-time decisions and uncertainty.

- After AI's victory in Go, attention turned to real-time strategy (RTS) games.
- RTS games were considered significantly more challenging due to:
 - Real-time control of multiple units
 - Decision-making under uncertainty
 - Long time horizons

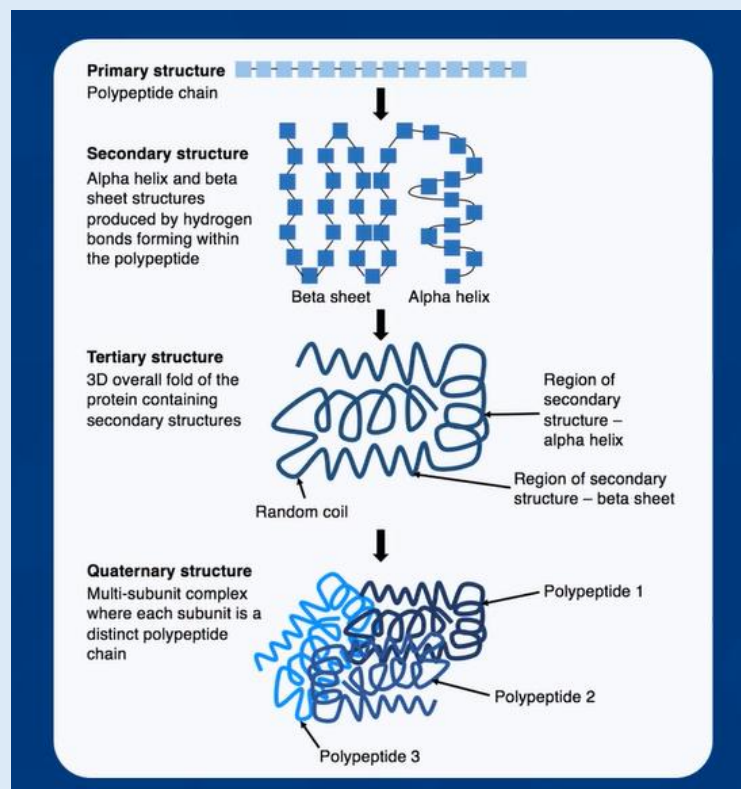
DOTA 2

OpenAI 5 and DOTA 2, 2019

STAR CRAFT

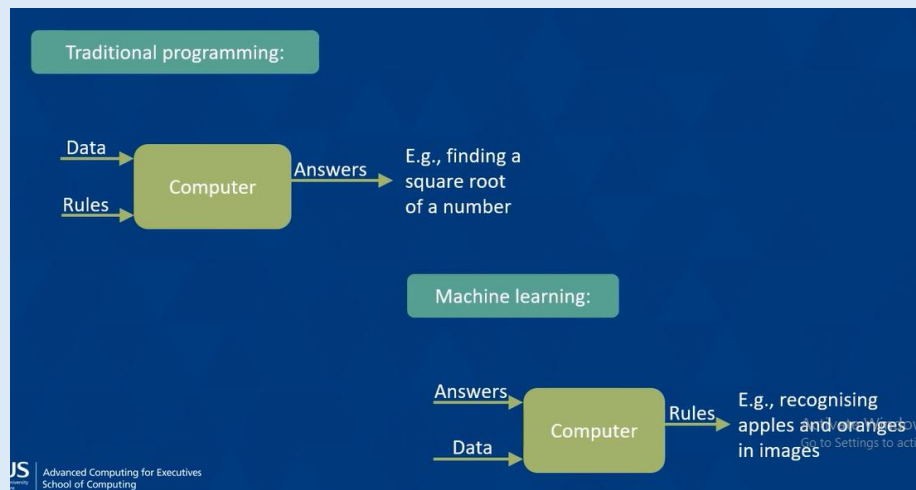
AlphaStar and Starcraft 2, 2019

- AlphaFold and ChatGPT showcase AI's impact beyond games, transforming biological research and enabling advanced human-like interactions.



What are the Types of ML Algorithms?

- Machine learning is an AI approach that enables computers to learn patterns and make decisions from data, much like human learning from experience.
- Traditional programming requires manual rule definition, while machine learning discovers these rules automatically from labelled input-output data.



- Visual classification tasks, like distinguishing apples from oranges, highlight the limitations of hardcoding and the strengths of data-driven learning.



- Machine learning identifies relationships, such as mathematical patterns between variables, through observation and generalisation, like in linear regression.

Can you find the relationship between x and y ?

x	-1	0	1	2	3	4	5	6	7
y	-4	1	6	11	16	21	26	31	36

$y = 5x + 1$

- Libraries like scikit-learn simplify machine learning implementation by providing tools to model, train, and make predictions using structured data.

```
import numpy as np
from sklearn.linear_model import LinearRegression

xvals = np.array([-1, 0, 1, 2, 3, 4, 5, 6, 7]).reshape((-1, 1))
yvals = np.array([-4, 1, 6, 11, 16, 21, 26, 31, 36]).reshape((-1, 1))

model = LinearRegression()

model.fit(xvals, yvals)

newx = np.array([8, 9, 10]).reshape((-1, 1))

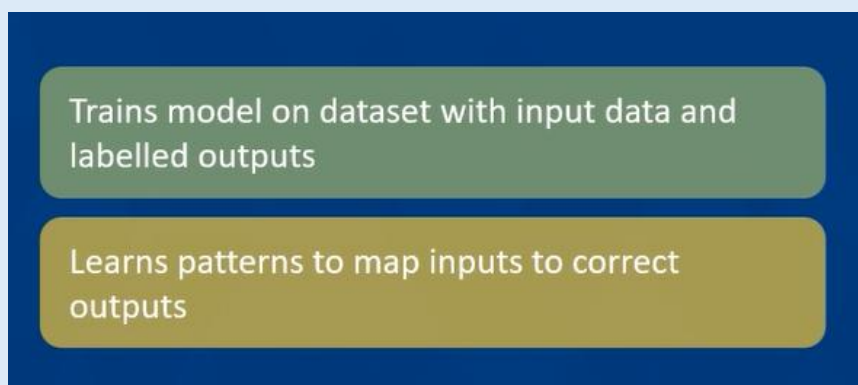
y_pred = model.predict(newx)

print('predicted y-values:', y_pred, sep='\n')
```

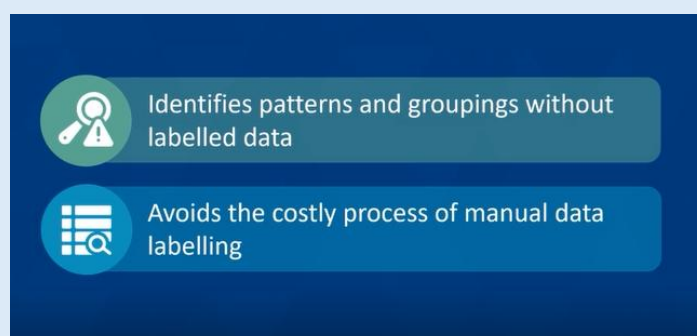
predicted y-values:
[[41.]
[46.]
[51.]]

What are the Types of ML Algorithms?

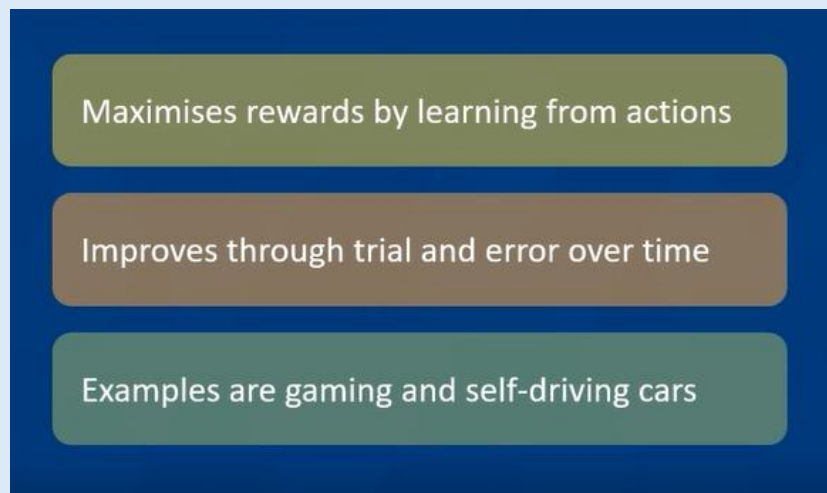
- Machine learning includes three major types: supervised learning, unsupervised learning, and reinforcement learning, each suited to different data and goals.
- Supervised learning trains models on labelled data to perform classification or regression, with steps including training, testing, and prediction.



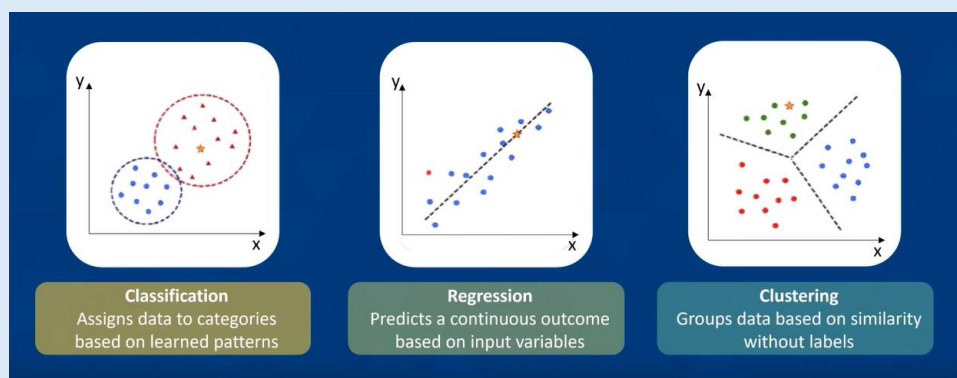
- Unsupervised learning works with unlabelled data to discover patterns through clustering, dimensionality reduction, or anomaly detection.



- Reinforcement learning teaches agents through trial and error to make sequential decisions by maximising rewards from an environment.

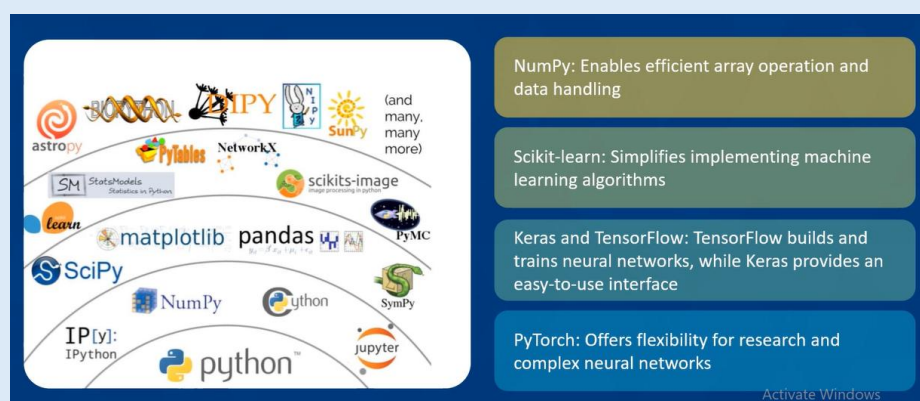


- Core ML tasks include classification, regression, and clustering, which help solve problems from predicting outcomes to discovering hidden data structures.

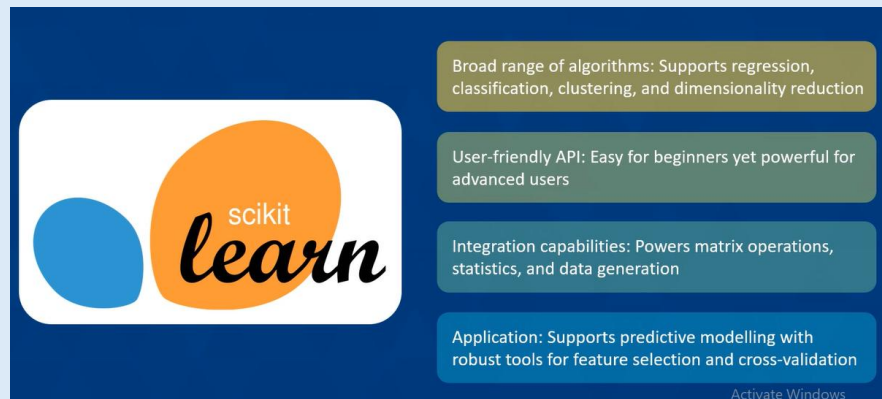


Overview of Python Libraries for ML and DL

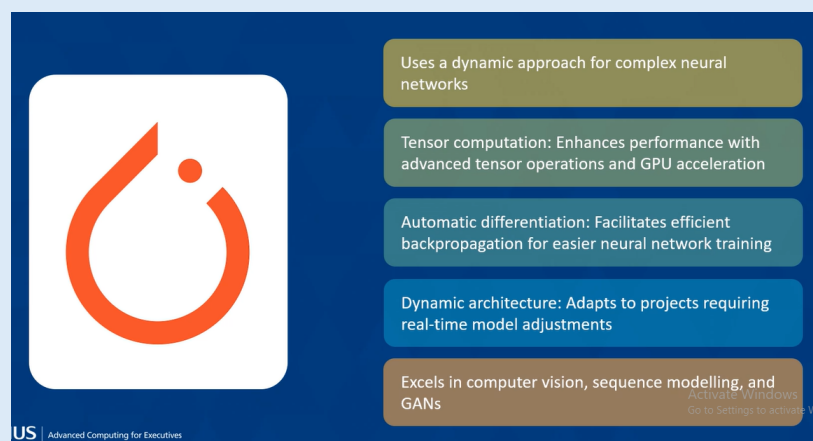
- Python's popularity in machine learning is driven by powerful libraries like NumPy, scikit-learn, TensorFlow, Keras, and PyTorch, supporting tasks from data handling to neural network training.



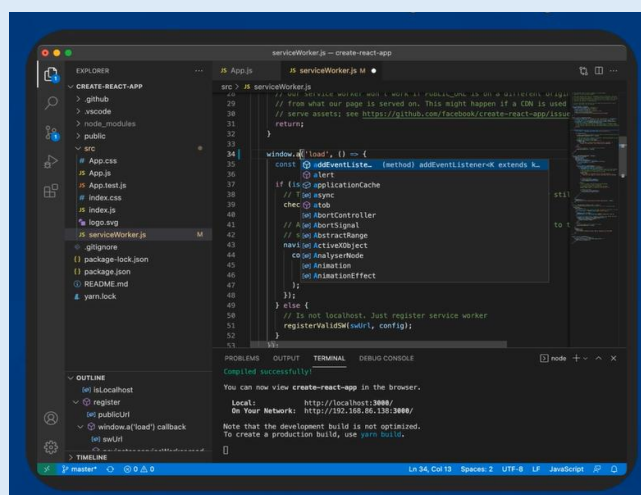
- NumPy enables efficient numerical computing with multi-dimensional arrays and matrix operations, forming the foundation for most ML and DL tasks.
- Scikit-learn simplifies classical machine learning with tools for regression, classification, clustering, and evaluation, suitable for both beginners and experts.



- TensorFlow, Keras, and PyTorch lead in deep learning, offering flexibility, GPU support, and ease of model building for tasks like image recognition and NLP.



- Popular Python IDEs such as Google Colab, Jupyter Notebook, VS Code, PyCharm, and Spyder enhance productivity with features like live code execution, debugging, and GPU integration.



Module Summary

- Artificial Intelligence is a multidisciplinary field focused on enabling machines to perform tasks that normally require human intelligence.
- Core branches of AI include machine learning, natural language processing, computer vision, and robotics, each addressing specific real-world challenges.



Key branches include machine learning, NLP, robotics, and computer vision.

- Landmark achievements like Deep Blue, AlphaGo, autonomous vehicles, and ChatGPT demonstrate AI's rapid progress and transformative impact.



Milestones include Deep Blue, AlphaGo, autonomous vehicles, and ChatGPT.

- Machine learning empowers systems to learn from data, with supervised, unsupervised, and reinforcement learning covering varied use cases.
- Python libraries such as NumPy, Scikit-learn, TensorFlow, and PyTorch support end-to-end AI development, from basic data handling to deep learning.



Key Python libraries like NumPy, Scikit-learn, TensorFlow, and PyTorch support ML and DL.

THE END